

Sub-Saharan Africa Diabetes Burden Analysis

GBD 2023 regional and subregional trends, forecasts, and burden composition

July 11, 2026

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1 Data Readiness

Table 1: Available inputs and enabled analyses

Input	Status	Enabled analysis
Age-standardized rates	AVAILABLE	Endpoints, trends, sex ratios, forecasts
All-age counts	AVAILABLE	Count endpoints, diabetes-type shares, YLL-to-YLD ratios
Age-specific counts	AVAILABLE	Age-specific count validation and readiness for decomposition
Age-specific rates	AVAILABLE	GBD-consistent decomposition, rate-effect estimation, age-period-cohort analysis
UN WPP 2024 population	AVAILABLE	Demographic-only projections and population sensitivity analysis

Notes: Age-standardized rates support comparisons across locations with different age structures. UN WPP population data can support demographic-only projections. Age-specific GBD rates are available and validated for GBD-consistent rate-effect decomposition and age-period-cohort analyses.

The report presents descriptive, trend, forecast, sex-difference, diabetes-type, and mortality-to-disability analyses for overall Sub-Saharan Africa and its four GBD subregions. Matched GBD age-specific rates and counts are now available for decomposition and age-period-cohort implementation. Existing UN WPP population data support demographic projection scenarios.

2 Key Findings

For overall SSA, age-standardized diabetes prevalence increased from 3543.5 to 4521.4 per 100,000 between 1990 and 2023, a 27.6% increase. Diabetes mortality increased from 33.2 to 38.3 per 100,000, a 15.3% increase.

Among the five regions, the highest 2023 age-standardized prevalence was in Southern SSA at 6626.6 per 100,000. The fastest DALY rate growth was in Southern SSA, with a 44.1% increase from 1990 to 2023. The overall SSA YLL-to-YLD ratio changed from 2.54 in 1990 to 1.94 in 2023, indicating the evolving balance between premature mortality and disability.

3 Measures and Interpretation

This analysis uses six GBD measures. They answer related but distinct questions, so they should not be interpreted interchangeably.

Table 2: GBD diabetes burden measures used in this analysis

Measure	What it measures	Main interpretation
Prevalence	Existing diabetes cases in a population at a given time.	How widespread diabetes is.
Incidence	New diabetes cases occurring during a year.	How quickly new diabetes burden is being generated.
Deaths	Deaths attributed to diabetes during a year.	Fatal burden.
YLLs	Years of life lost because diabetes deaths occurred before the reference life expectancy.	Premature-mortality burden.
YLDs	Years lived with disability from diabetes and its sequelae.	Non-fatal disability burden.
DALYs	Total health loss: YLLs plus YLDs.	Combined fatal and non-fatal burden.

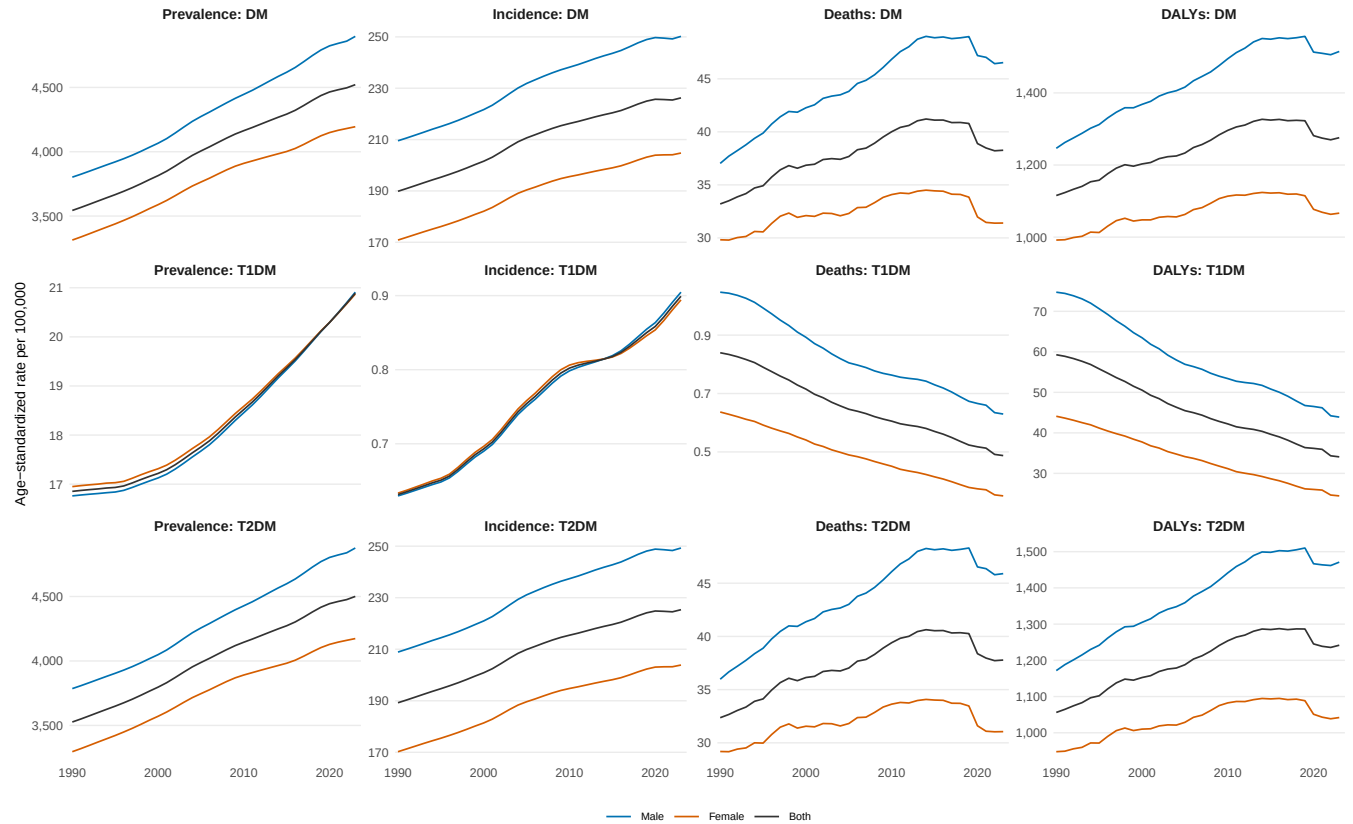
Notes: Age-standardized rates are used for comparisons across regions. Counts measure absolute burden and are affected by population size and age structure.

4 Overall SSA Results

Table 3: Age-standardized rates, trends, and forecasts for overall SSA

Measure	Type	1990 rate	2023 rate	Change, %	APC, % (95% CI)	Model	2030 forecast (95% PI)	p
Prevalence	DM	3543.5	4521.4	27.6	0.78 (0.76-0.79)	ARIMA	4740.5 (4687.2-4793.8)	<0.001
Prevalence	T1DM	16.9	20.9	23.9	0.69 (0.64-0.75)	ETS	22.3 (22.0-22.6)	<0.001
Prevalence	T2DM	3526.7	4500.5	27.6	0.78 (0.76-0.80)	ARIMA	4718.7 (4665.6-4771.8)	<0.001
Incidence	DM	189.9	226.2	19.2	0.57 (0.54-0.60)	ARIMA	237.4 (227.3-247.4)	<0.001
Incidence	T1DM	0.6	0.9	42.4	1.11 (1.05-1.16)	ARIMA	1.0 (0.9-1.0)	<0.001
Incidence	T2DM	189.2	225.3	19.1	0.57 (0.54-0.60)	ARIMA	236.3 (226.3-246.3)	<0.001
Deaths	DM	33.2	38.3	15.3	0.56 (0.43-0.68)	ETS	36.2 (32.1-40.3)	<0.001
Deaths	T1DM	0.8	0.5	-41.9	-1.64 (-1.68-1.59)	ARIMA	0.4 (0.4-0.4)	<0.001
Deaths	T2DM	32.4	37.8	16.7	0.60 (0.47-0.72)	ETS	35.8 (31.5-40.0)	<0.001
DALYs	DM	1115.4	1275.8	14.4	0.50 (0.41-0.58)	ETS	1245.3 (1150.6-1340.0)	<0.001
DALYs	T1DM	59.3	34.1	-42.5	-1.69 (-1.73-1.65)	ETS	29.3 (26.0-32.6)	<0.001
DALYs	T2DM	1056.1	1241.7	17.6	0.58 (0.50-0.67)	ETS	1215.9 (1120.9-1310.9)	<0.001

Notes: Rates are age-standardized GBD estimates per 100,000 population. APC is the average annual percentage change from a log-linear model. PI is the model-based prediction interval for the 2030 forecast. DALYs are disability-adjusted life years, combining fatal burden and non-fatal disability.



Source: GBD 2023. Lines show point estimates.

Figure 1: Overall SSA diabetes burden by measure, diabetes type, and sex, 1990-2023.

5 Overall SSA Extensions

Table 4: Overall SSA age-standardized endpoint rates

Measure	1990 rate (95% UI)	2023 rate (95% UI)	Change, %
Prevalence	3543.5 (3211.6-3861.8)	4521.4 (4145.1-4912.2)	27.6
Incidence	189.9 (174.0-206.3)	226.2 (209.5-244.5)	19.2
Deaths	33.2 (24.7-43.0)	38.3 (29.1-49.7)	15.3
DALYs	1115.4 (891.8-1378.7)	1275.8 (1027.3-1533.4)	14.4
YLLs	787.2 (585.8-1021.4)	858.7 (660.3-1103.6)	9.1
YLDs	328.2 (228.7-443.6)	417.1 (290.1-561.8)	27.1

Notes: UI is the GBD 95 percent uncertainty interval. YLLs are years of life lost from premature mortality; YLDs are years lived with disability. DALYs are the sum of YLLs and YLDs.

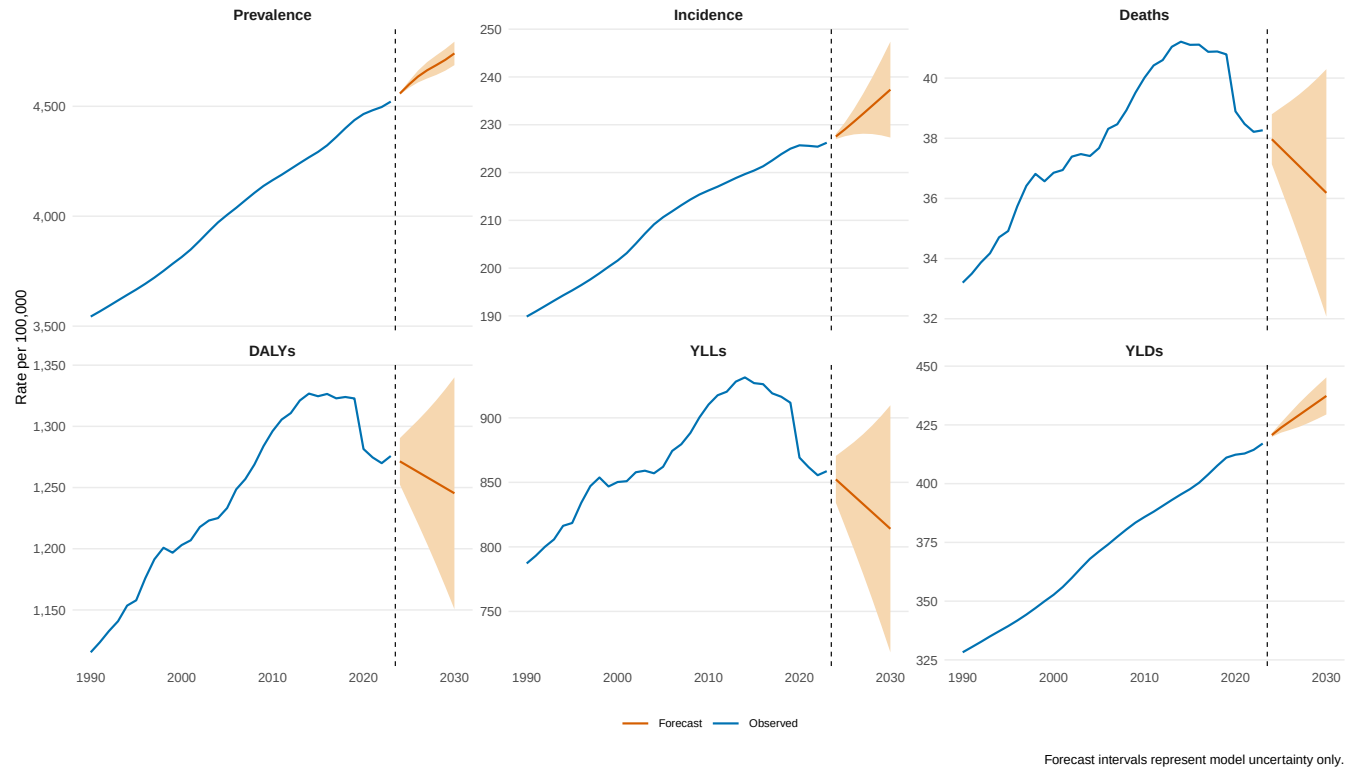


Figure 2: Overall SSA observed and forecast age-standardized rates through 2030.

Table 5: Overall SSA all-age count endpoints

Measure	1990 count (95% UI)	2023 count (95% UI)	Change, %
Prevalence	9489400 (8575992-10390241)	31820019 (28916708-34583822)	235.3
Incidence	557276 (509485-603342)	1783394 (1636186-1931284)	220.0
Deaths	67947 (50638-87745)	190336 (145578-245515)	180.1
DALYs	2903535 (2328463-3562374)	7998941 (6452703-9622946)	175.5
YLLs	2082249 (1590962-2647966)	5277446 (4073393-6796153)	153.4
YLDs	821287 (568273-1120571)	2721496 (1886373-3666835)	231.4

Notes: Counts are all-age numbers, not age-standardized rates. YLLs and YLDs are time-equivalent health loss measures; DALYs equal YLLs plus YLDs.

6 Overall Region Versus Subregions



Figure 3: Age-standardized total diabetes trends in overall SSA and four subregions.

Table 6: Overall SSA and subregional age-standardized endpoint comparison

Region	Measure	1990 rate	2023 rate	Change, %
SSA	Prevalence	3543.5	4521.4	27.6
SSA	Incidence	189.9	226.2	19.2
SSA	Deaths	33.2	38.3	15.3
SSA	DALYs	1115.4	1275.8	14.4
SSA	YLLs	787.2	858.7	9.1
SSA	YLDs	328.2	417.1	27.1
Central SSA	Prevalence	5009.6	6217.1	24.1
Central SSA	Incidence	242.1	284.7	17.6
Central SSA	Deaths	30.6	33.9	11.0
Central SSA	DALYs	1207.2	1384.0	14.6
Central SSA	YLLs	751.1	819.6	9.1
Central SSA	YLDs	456.1	564.4	23.7
Eastern SSA	Prevalence	2717.4	3142.8	15.7
Eastern SSA	Incidence	159.7	173.8	8.8
Eastern SSA	Deaths	36.7	36.3	-1.3
Eastern SSA	DALYs	1163.7	1135.0	-2.5
Eastern SSA	YLLs	898.4	828.0	-7.8
Eastern SSA	YLDs	265.3	306.9	15.7
Southern SSA	Prevalence	5191.0	6626.6	27.7
Southern SSA	Incidence	288.9	359.2	24.4
Southern SSA	Deaths	40.2	63.0	56.9
Southern SSA	DALYs	1391.9	2005.3	44.1
Southern SSA	YLLs	900.4	1378.8	53.1
Southern SSA	YLDs	491.5	626.5	27.5
Western SSA	Prevalence	3364.7	4604.0	36.8
Western SSA	Incidence	172.7	218.0	26.2
Western SSA	Deaths	28.2	32.6	15.6
Western SSA	DALYs	952.3	1122.9	17.9
Western SSA	YLLs	654.5	716.3	9.4
Western SSA	YLDs	297.7	406.6	36.6

Notes: Rates are per 100,000 population and age-standardized. Change is the percentage change between 1990 and 2023.

Table 7: Overall SSA and subregional full-period log-linear trends

Region	Measure	APC, % (95% CI)	p	Adj. R2
SSA	Prevalence	0.78 (0.76-0.79)	<0.001	0.996
SSA	Incidence	0.57 (0.54-0.60)	<0.001	0.978
SSA	Deaths	0.56 (0.43-0.68)	<0.001	0.720
SSA	DALYs	0.50 (0.41-0.58)	<0.001	0.818
SSA	YLLs	0.38 (0.26-0.49)	<0.001	0.575
SSA	YLDs	0.77 (0.75-0.79)	<0.001	0.993
Central SSA	Prevalence	0.66 (0.64-0.68)	<0.001	0.993
Central SSA	Incidence	0.47 (0.45-0.49)	<0.001	0.985
Central SSA	Deaths	0.61 (0.51-0.71)	<0.001	0.823
Central SSA	DALYs	0.60 (0.53-0.66)	<0.001	0.910
Central SSA	YLLs	0.56 (0.45-0.67)	<0.001	0.763
Central SSA	YLDs	0.65 (0.63-0.67)	<0.001	0.994
Eastern SSA	Prevalence	0.45 (0.44-0.45)	<0.001	0.998
Eastern SSA	Incidence	0.28 (0.27-0.29)	<0.001	0.986
Eastern SSA	Deaths	0.06 (-0.01-0.12)	0.094	0.057
Eastern SSA	DALYs	-0.03 (-0.07-0.01)	0.193	0.023
Eastern SSA	YLLs	-0.18 (-0.24-0.13)	<0.001	0.556
Eastern SSA	YLDs	0.46 (0.45-0.47)	<0.001	0.997
Southern SSA	Prevalence	0.90 (0.85-0.94)	<0.001	0.979
Southern SSA	Incidence	0.81 (0.75-0.87)	<0.001	0.958
Southern SSA	Deaths	1.35 (0.98-1.71)	<0.001	0.627
Southern SSA	DALYs	1.16 (0.90-1.42)	<0.001	0.709
Southern SSA	YLLs	1.28 (0.90-1.65)	<0.001	0.592
Southern SSA	YLDs	0.90 (0.85-0.96)	<0.001	0.973
Western SSA	Prevalence	0.98 (0.96-1.00)	<0.001	0.996
Western SSA	Incidence	0.74 (0.70-0.77)	<0.001	0.983
Western SSA	Deaths	0.63 (0.53-0.74)	<0.001	0.823
Western SSA	DALYs	0.62 (0.55-0.69)	<0.001	0.909
Western SSA	YLLs	0.45 (0.35-0.54)	<0.001	0.738
Western SSA	YLDs	0.98 (0.95-1.01)	<0.001	0.993

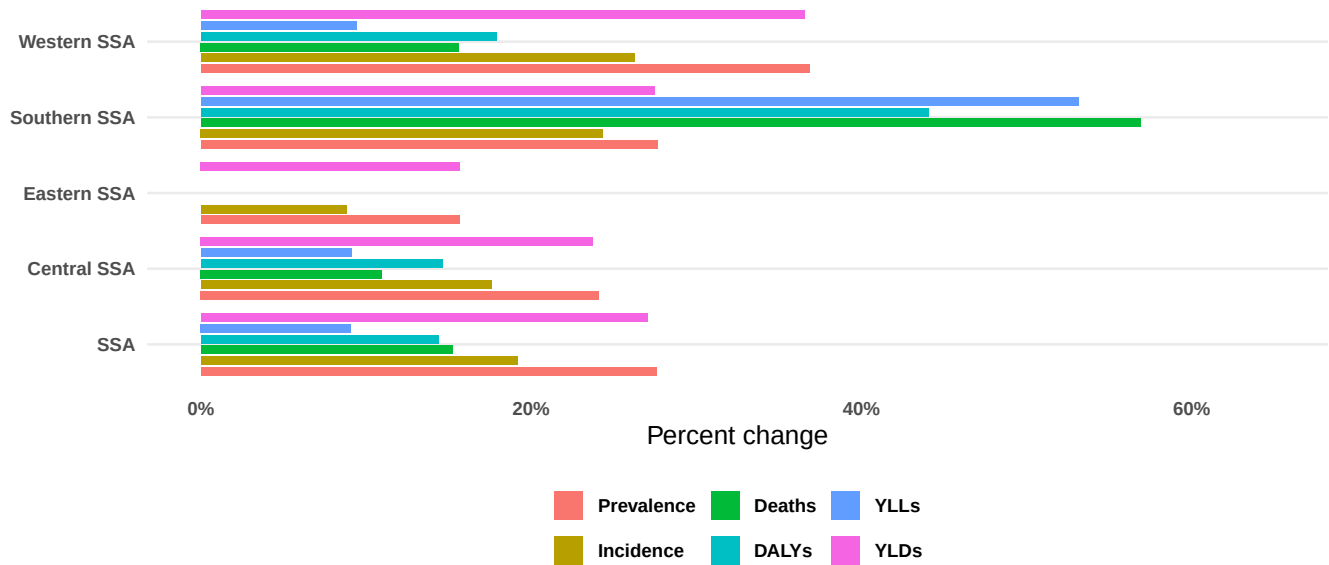
Notes: APC is the average annual percentage change from a log-linear model across 1990-2023. Adj. R2 is the adjusted coefficient of determination for that trend model.

Table 8: Subregional rate ratios relative to overall SSA

Region	Measure	Year	Rate ratio vs SSA
Central SSA	Deaths	1990	0.920
Central SSA	DALYs	1990	1.082
Central SSA	YLDs	1990	1.390
Central SSA	YLLs	1990	0.954
Central SSA	Prevalence	1990	1.414
Central SSA	Incidence	1990	1.275
Central SSA	Deaths	2023	0.886
Central SSA	DALYs	2023	1.085
Central SSA	YLDs	2023	1.353
Central SSA	YLLs	2023	0.954
Central SSA	Prevalence	2023	1.375
Central SSA	Incidence	2023	1.258
Eastern SSA	Deaths	1990	1.106
Eastern SSA	DALYs	1990	1.043
Eastern SSA	YLDs	1990	0.808
Eastern SSA	YLLs	1990	1.141
Eastern SSA	Prevalence	1990	0.767
Eastern SSA	Incidence	1990	0.841
Eastern SSA	Deaths	2023	0.948
Eastern SSA	DALYs	2023	0.890
Eastern SSA	YLDs	2023	0.736
Eastern SSA	YLLs	2023	0.964
Eastern SSA	Prevalence	2023	0.695
Eastern SSA	Incidence	2023	0.768
Southern SSA	Deaths	1990	1.210
Southern SSA	DALYs	1990	1.248
Southern SSA	YLDs	1990	1.497
Southern SSA	YLLs	1990	1.144
Southern SSA	Prevalence	1990	1.465
Southern SSA	Incidence	1990	1.521
Southern SSA	Deaths	2023	1.647
Southern SSA	DALYs	2023	1.572
Southern SSA	YLDs	2023	1.502
Southern SSA	YLLs	2023	1.606
Southern SSA	Prevalence	2023	1.466
Southern SSA	Incidence	2023	1.588
Western SSA	Deaths	1990	0.850
Western SSA	DALYs	1990	0.854
Western SSA	YLDs	1990	0.907
Western SSA	YLLs	1990	0.832

Western SSA	Prevalence	1990	0.950
Western SSA	Incidence	1990	0.910
Western SSA	Deaths	2023	0.853
Western SSA	DALYs	2023	0.880
Western SSA	YLDs	2023	0.975
Western SSA	YLLs	2023	0.834
Western SSA	Prevalence	2023	1.018
Western SSA	Incidence	2023	0.964

Notes: A ratio above 1 means the subregion has a higher age-standardized rate than overall SSA in that year. A ratio below 1 means the subregion has a lower rate than overall SSA.



Both sexes, total diabetes mellitus. Source: GBD 2023.

Figure 4: Percent change in total diabetes age-standardized rates, 1990-2023.

7 Sex Differences and Diabetes Type Mix

Table 9: Male-to-female age-standardized rate ratios in 2023

Region	Measure	Male	Female	M:F ratio
SSA	Prevalence	4896.9	4194.5	1.2
SSA	Incidence	250.2	204.8	1.2
SSA	Deaths	46.5	31.4	1.5
SSA	DALYs	1514.9	1066.5	1.4
SSA	YLLs	1058.4	683.6	1.5
SSA	YLDs	456.5	382.9	1.2
Central SSA	Prevalence	7095.8	5461.9	1.3
Central SSA	Incidence	328.2	244.5	1.3
Central SSA	Deaths	48.2	23.4	2.1
Central SSA	DALYs	1804.4	1034.2	1.7
Central SSA	YLLs	1154.4	542.0	2.1
Central SSA	YLDs	650.0	492.2	1.3
Eastern SSA	Prevalence	3351.9	2963.2	1.1
Eastern SSA	Incidence	194.5	154.8	1.3
Eastern SSA	Deaths	44.6	29.0	1.5
Eastern SSA	DALYs	1354.2	935.8	1.4
Eastern SSA	YLLs	1023.3	649.5	1.6
Eastern SSA	YLDs	330.8	286.3	1.2
Southern SSA	Prevalence	7484.5	5950.1	1.3
Southern SSA	Incidence	395.4	327.0	1.2
Southern SSA	Deaths	68.8	59.1	1.2
Southern SSA	DALYs	2230.6	1836.6	1.2
Southern SSA	YLLs	1518.1	1276.4	1.2
Southern SSA	YLDs	712.6	560.2	1.3
Western SSA	Prevalence	4941.3	4301.5	1.1
Western SSA	Incidence	238.7	199.4	1.2
Western SSA	Deaths	41.1	25.2	1.6
Western SSA	DALYs	1361.3	908.9	1.5
Western SSA	YLLs	918.6	534.8	1.7
Western SSA	YLDs	442.7	374.1	1.2

Notes: M:F ratio is the male age-standardized rate divided by the female age-standardized rate.

Table 10: Diabetes type shares of all-age burden in 2023

Region	Measure	T1DM share, %	T2DM share, %	Component difference, %
SSA	Prevalence	0.7	99.3	0
SSA	Incidence	0.7	99.3	0
SSA	Deaths	3.5	96.5	0
SSA	DALYs	6.3	93.7	0
SSA	YLLs	9.2	90.8	0
SSA	YLDs	0.6	99.4	0
Central SSA	Prevalence	0.4	99.6	0
Central SSA	Incidence	0.4	99.6	0
Central SSA	Deaths	4.7	95.3	0
Central SSA	DALYs	6.1	93.9	0
Central SSA	YLLs	10.4	89.6	0
Central SSA	YLDs	0.3	99.7	0
Eastern SSA	Prevalence	1.6	98.4	0
Eastern SSA	Incidence	1.7	98.3	0
Eastern SSA	Deaths	4.3	95.7	0
Eastern SSA	DALYs	7.9	92.1	0
Eastern SSA	YLLs	10.4	89.6	0
Eastern SSA	YLDs	1.2	98.8	0
Southern SSA	Prevalence	0.3	99.7	0
Southern SSA	Incidence	0.2	99.8	0
Southern SSA	Deaths	0.9	99.1	0
Southern SSA	DALYs	1.6	98.4	0
Southern SSA	YLLs	2.3	97.7	0
Southern SSA	YLDs	0.3	99.7	0
Western SSA	Prevalence	0.4	99.6	0
Western SSA	Incidence	0.4	99.6	0
Western SSA	Deaths	3.9	96.1	0
Western SSA	DALYs	7.1	92.9	0
Western SSA	YLLs	11.1	88.9	0
Western SSA	YLDs	0.4	99.6	0

Notes: T1DM is type 1 diabetes mellitus; T2DM is type 2 diabetes mellitus. Component difference shows the percent difference between T1DM plus T2DM and total diabetes; small nonzero values reflect GBD estimation and rounding.

8 Mortality-to-Disability Transition

```
## # A tibble: 6 x 5
##   location_name      year yll_count yld_count yll_to_yld_ratio
##   <chr>          <dbl>   <dbl>   <dbl>         <dbl>
## 1 Central Sub-Saharan Africa 1990   217150.  127191.         1.71
## 2 Central Sub-Saharan Africa 1991   224366.  131710.         1.70
## 3 Central Sub-Saharan Africa 1992   229655.  136462.         1.68
## 4 Central Sub-Saharan Africa 1993   234299.  141307.         1.66
## 5 Central Sub-Saharan Africa 1994   238291.  146323.         1.63
## 6 Central Sub-Saharan Africa 1995   243077.  151324.         1.61
```

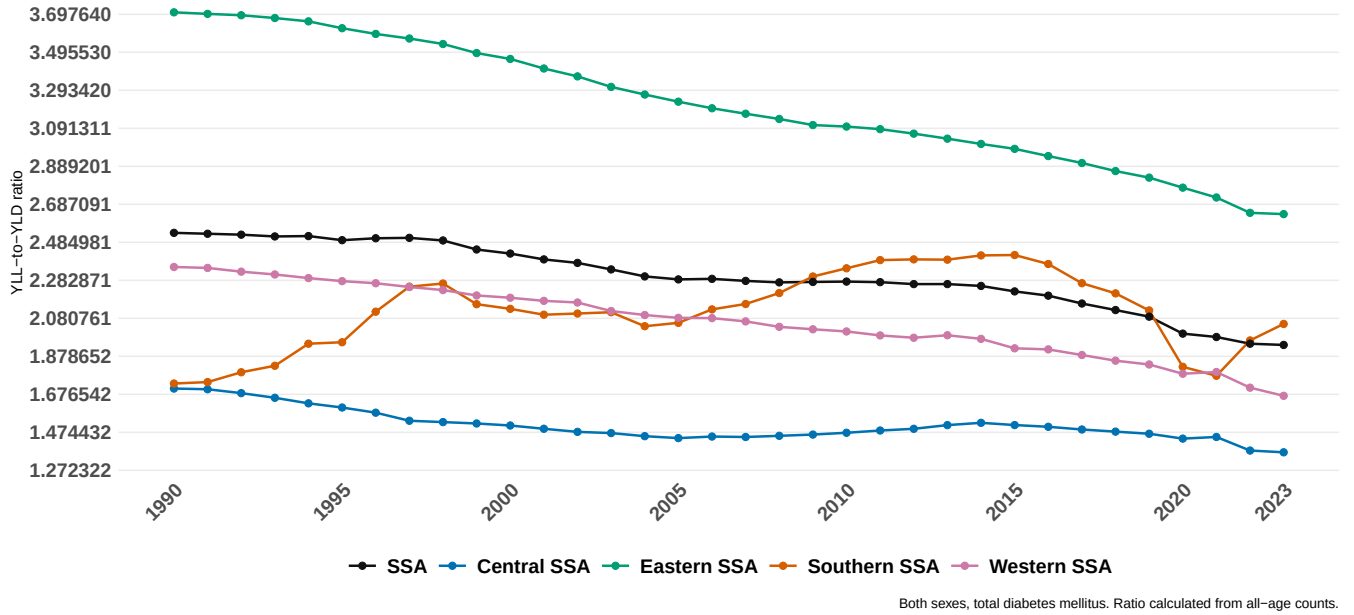


Figure 5: YLL-to-YLD ratio in overall SSA and subregions, 1990-2023.

Table 11: Period mean YLL-to-YLD ratios

Region	Period	YLL:YLD
Central SSA	1990-1994	1.676
Central SSA	1995-1999	1.554
Central SSA	2000-2004	1.481
Central SSA	2005-2009	1.453
Central SSA	2010-2014	1.497
Central SSA	2015-2019	1.490
Central SSA	2020-2023	1.409
Eastern SSA	1990-1994	3.688
Eastern SSA	1995-1999	3.564
Eastern SSA	2000-2004	3.364
Eastern SSA	2005-2009	3.170
Eastern SSA	2010-2014	3.059
Eastern SSA	2015-2019	2.905
Eastern SSA	2020-2023	2.694
Southern SSA	1990-1994	1.809
Southern SSA	1995-1999	2.148
Southern SSA	2000-2004	2.098
Southern SSA	2005-2009	2.172
Southern SSA	2010-2014	2.388
Southern SSA	2015-2019	2.278
Southern SSA	2020-2023	1.903
SSA	1990-1994	2.525
SSA	1995-1999	2.491
SSA	2000-2004	2.368
SSA	2005-2009	2.281
SSA	2010-2014	2.266
SSA	2015-2019	2.160

SSA	2020-2023	1.966
Western SSA	1990-1994	2.328
Western SSA	1995-1999	2.245
Western SSA	2000-2004	2.149
Western SSA	2005-2009	2.057
Western SSA	2010-2014	1.988
Western SSA	2015-2019	1.883
Western SSA	2020-2023	1.740

Notes: Higher YLL:YLD ratios indicate a larger share of burden from premature mortality relative to disability.

9 Pre-Forecast Diagnostics

Before fitting forecasting models, each overall SSA total-diabetes rate series was inspected for smoothing behavior, autocorrelation structure, partial autocorrelation structure, and differencing needs. These diagnostics are shown for the six core GBD measures because they are the headline forecast outcomes.

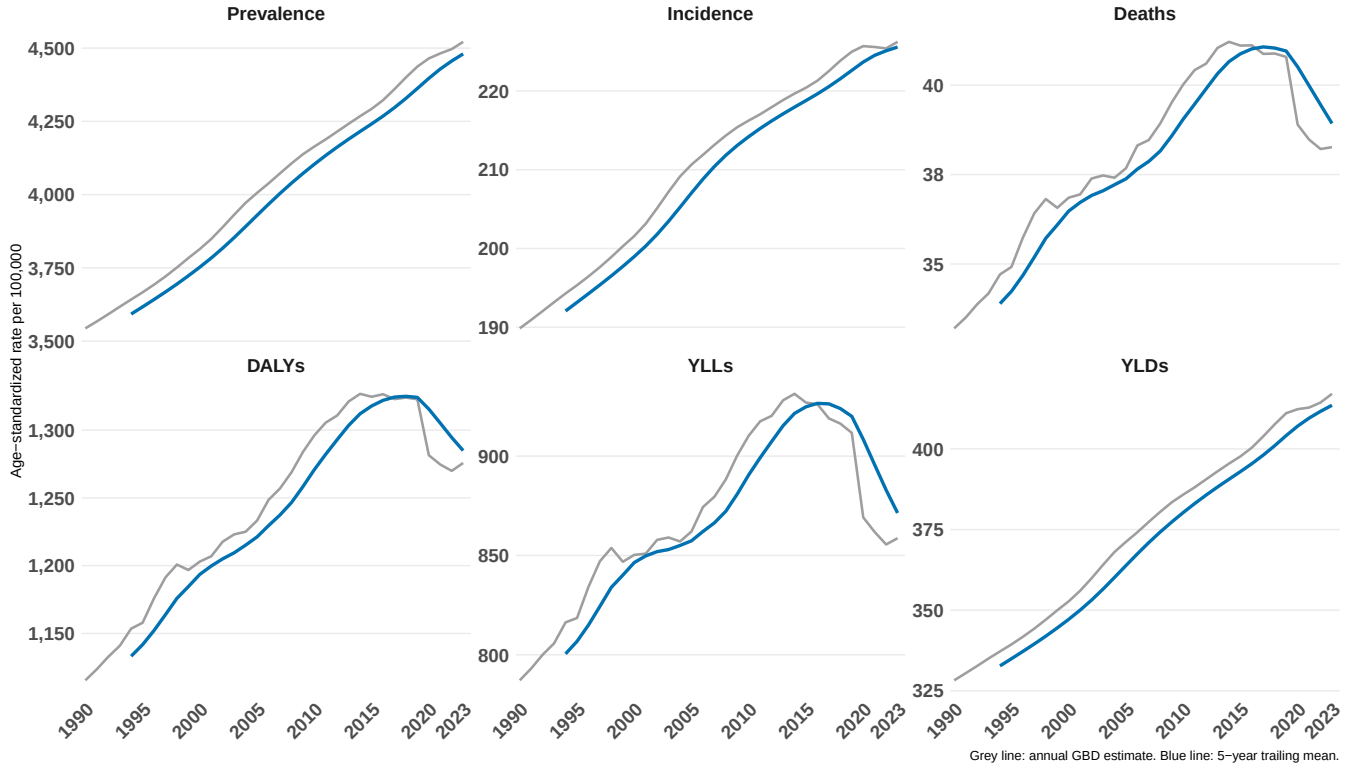


Figure 6: Observed rates and 5-year trailing means for overall SSA, total diabetes.

Table 12: Stationarity and ARIMA differencing diagnostics for overall SSA total diabetes rates

Measure	KPSS ndiffs	ADF ndiffs	PP ndiffs	ARIMA p	ARIMA d	ARIMA q	Selected ARIMA
DALYs	2	2	1	0	2	1	ARIMA(0,2,1)
Deaths	2	2	1	0	2	1	ARIMA(0,2,1)
Incidence	2	1	2	1	2	0	ARIMA(1,2,0)
Prevalence	1	1	2	3	1	0	ARIMA(3,1,0)
YLDs	1	1	2	0	1	3	ARIMA(0,1,3)
YLLs	2	2	1	0	2	1	ARIMA(0,2,1)

Notes: ndiffs is the number of differences suggested by each unit-root or stationarity test. KPSS tests stationarity as the null hypothesis; ADF and PP test a unit root as the null hypothesis. The selected ARIMA order is the order returned by `auto.arima` for the undifferenced input series.

In `auto.arima()`, the differencing order d is selected before searching over AR and MA orders. For a non-seasonal annual series, the function uses unit-root and stationarity tests through `ndiffs()` to estimate the smallest number of differences needed to make the series approximately stationary. In this report, KPSS, ADF, and PP suggestions are shown side by side, while the `auto_arima_d` column records the value actually used by `auto.arima()` in the selected model. After d is chosen, `auto.arima()` searches over candidate (p, d, q) models and chooses the specification with the best information criterion, subject to its default constraints.

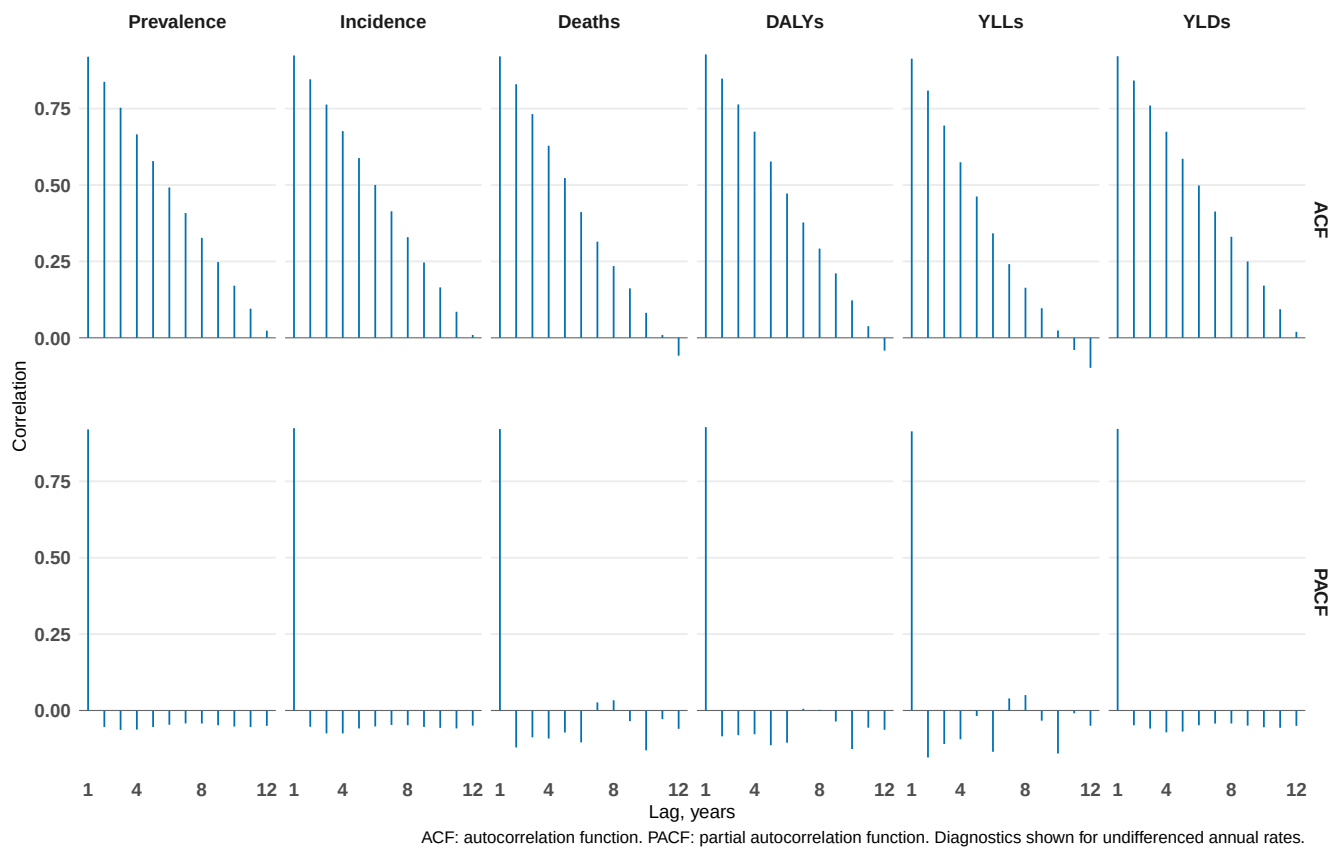


Figure 7: ACF and PACF diagnostics for overall SSA total diabetes rates.

10 Forecasting Models and Validation

The forecasting target is the annual age-standardized rate y_t for a single stratum defined by location, measure, diabetes cause, and sex. Calendar year t is the time index. Models are trained on observed GBD estimates from 1990 to 2023 and evaluated using expanding-window one-step-ahead validation. The initial training window ends in 2004; the model is then refit repeatedly to forecast each next year through 2023.

Table 13: Forecasting model registry

Model	Status	Role in analysis
Log-linear trend	Implemented	Average annual percentage change over 1990-2023
ARIMA	Implemented	Short-term age-standardized rate forecasting
ETS	Implemented	Short-term age-standardized rate forecasting
LSTM	Implemented and fitted for headline comparison	Machine-learning forecast candidate for nonlinear temporal patterns

Notes: LSTM fitting requires the R keras/tensorflow packages plus a working Python TensorFlow backend. TensorFlow backend available: 2.21

10.1 Model Specifications

Log-linear trend model. The descriptive trend model is

$$\log(y_t) = \alpha + \beta t + \epsilon_t.$$

The average annual percentage change is

$$APC = 100\{\exp(\beta) - 1\}.$$

This model is used for interpretable long-run change, not for selecting the best forecast.

ARIMA. The autoregressive integrated moving-average model is written as

$$\phi(B)(1 - B)^d y_t = c + \theta(B)\epsilon_t,$$

where B is the backshift operator, d is the differencing order, and $\phi(B)$ and $\theta(B)$ are autoregressive and moving-average polynomials. ARIMA is included because diabetes rates may contain serial dependence, gradual trend, and short-term autocorrelated deviations.

ETS. The exponential-smoothing state-space model decomposes the series into level, trend, and error components. In the additive-error form,

$$\begin{aligned} y_t &= \ell_{t-1} + b_{t-1} + \epsilon_t, \\ \ell_t &= \ell_{t-1} + b_{t-1} + \alpha\epsilon_t, \quad b_t = b_{t-1} + \beta\epsilon_t. \end{aligned}$$

ETS is included because it performs well for smooth epidemiologic trajectories where recent observations should receive more weight than distant observations.

LSTM. The long short-term memory model is specified as a recurrent neural network for lagged first-differenced rate sequences, $\Delta y_t = y_t - y_{t-1}$. The fitted network predicts $\widehat{\Delta y}_{t+1}$, and the rate forecast is reconstructed as $\hat{y}_{t+1} = y_t + \widehat{\Delta y}_{t+1}$. For a scaled input vector x_t , the core gates are

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f), \\ i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), \quad \tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o), \\ h_t &= o_t \odot \tanh(c_t). \end{aligned}$$

The forecast is generated from the hidden state h_t . LSTMs are theoretically attractive because the gated cell state can represent nonlinear temporal patterns and longer memory than standard autoregressive models. They are not assumed to outperform ARIMA automatically; performance must be judged by the validation metrics below.

10.2 Variables Used

All forecasting models use the same outcome: annual age-standardized rate per 100,000 population. Models are fit separately for each location, measure, diabetes cause, and sex. No external covariates are included in the current forecasting stage; therefore, forecasts represent continuation of historical rate patterns, not causal projections under policy or exposure scenarios.

10.3 Model Comparison Metrics

For validation year t , let y_t be the observed rate and $\hat{y}_t$ the one-step-ahead forecast. With n validation years:

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2},$$

$$MAE = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|,$$

$$MAPE = \frac{100}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right|,$$

$$MASE = \frac{\frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|}{\frac{1}{T-1} \sum_{t=2}^T |y_t - y_{t-1}|}.$$

Lower values indicate better validation performance. MASE is the primary selection metric because it is scale-free and compares model error with a naive random-walk benchmark.

For LSTM forecasts, 95 percent prediction intervals are estimated from the empirical distribution of expanding-window validation residuals. The interval is centered on the recursive LSTM point forecast and widened by the square root of the forecast horizon. These intervals are therefore validation-residual intervals rather than parametric state-space intervals.

10.4 Model Comparison Results

The production 2030 forecast is selected from ARIMA and ETS for every location-measure-cause-sex stratum because those models provide stable prediction intervals for all series. The headline model comparison separately fits ARIMA, ETS, and LSTM for overall SSA, both sexes, total diabetes across all six measures. These headline results are used to compare machine-learning validation performance against classical time-series models.

Table 14: Headline forecast model comparison for overall SSA total diabetes rates

Measure	Model	RMSE	MAE	MAPE	MASE	Validation years	Selected	Status
Prevalence	ARIMA	3.804	3.079	0.070	0.102	9	Yes	Fitted
Prevalence	ETS	6.728	5.761	0.130	0.190	9	No	Fitted
Prevalence	LSTM	9.140	7.807	0.176	0.258	9	No	Fitted
Incidence	ARIMA	0.360	0.285	0.126	0.237	9	Yes	Fitted
Incidence	ETS	0.478	0.350	0.156	0.292	9	No	Fitted
Incidence	LSTM	0.566	0.410	0.182	0.342	9	No	Fitted
Deaths	ARIMA	0.790	0.596	1.511	1.724	9	No	Fitted
Deaths	ETS	0.735	0.536	1.364	1.552	9	No	Fitted
Deaths	LSTM	0.727	0.425	1.074	1.232	9	Yes	Fitted
DALYs	ARIMA	20.639	15.567	1.207	1.799	9	No	Fitted
DALYs	ETS	17.098	12.915	1.001	1.492	9	No	Fitted
DALYs	LSTM	15.855	10.267	0.792	1.186	9	Yes	Fitted
YLLs	ARIMA	20.232	15.399	1.752	2.261	9	No	Fitted
YLLs	ETS	16.757	12.882	1.467	1.892	9	No	Fitted
YLLs	LSTM	15.300	10.032	1.132	1.473	9	Yes	Fitted
YLDs	ARIMA	0.764	0.527	0.128	0.189	9	Yes	Fitted
YLDs	ETS	0.991	0.787	0.191	0.282	9	No	Fitted
YLDs	LSTM	1.193	0.909	0.221	0.326	9	No	Fitted

Notes: The headline comparison uses expanding-window one-step-ahead validation. LSTM is fitted to lagged first-differenced rates and transformed back to the original rate scale for validation. The selected model is the one with the lowest MASE within each measure.

Table 15: Selected headline forecast model by measure

Measure	Selected headline model	RMSE	MAE	MAPE	MASE
Prevalence	ARIMA	3.804	3.079	0.070	0.102
Incidence	ARIMA	0.360	0.285	0.126	0.237
Deaths	LSTM	0.727	0.425	1.074	1.232
DALYs	LSTM	15.855	10.267	0.792	1.186
YLLs	LSTM	15.300	10.032	1.132	1.473
YLDs	ARIMA	0.764	0.527	0.128	0.189

Notes: Selection is based on the lowest MASE among ARIMA, ETS, and LSTM. This table applies to overall SSA, both sexes, total diabetes.

Table 16: Production forecast model selection across all strata

Model	Selected strata	Share selected, %
ARIMA	173	64.1
ETS	97	35.9

Notes: A stratum is one unique location, measure, diabetes cause, and sex combination. This production table compares ARIMA and ETS for all strata; LSTM results are shown in the headline comparison above.

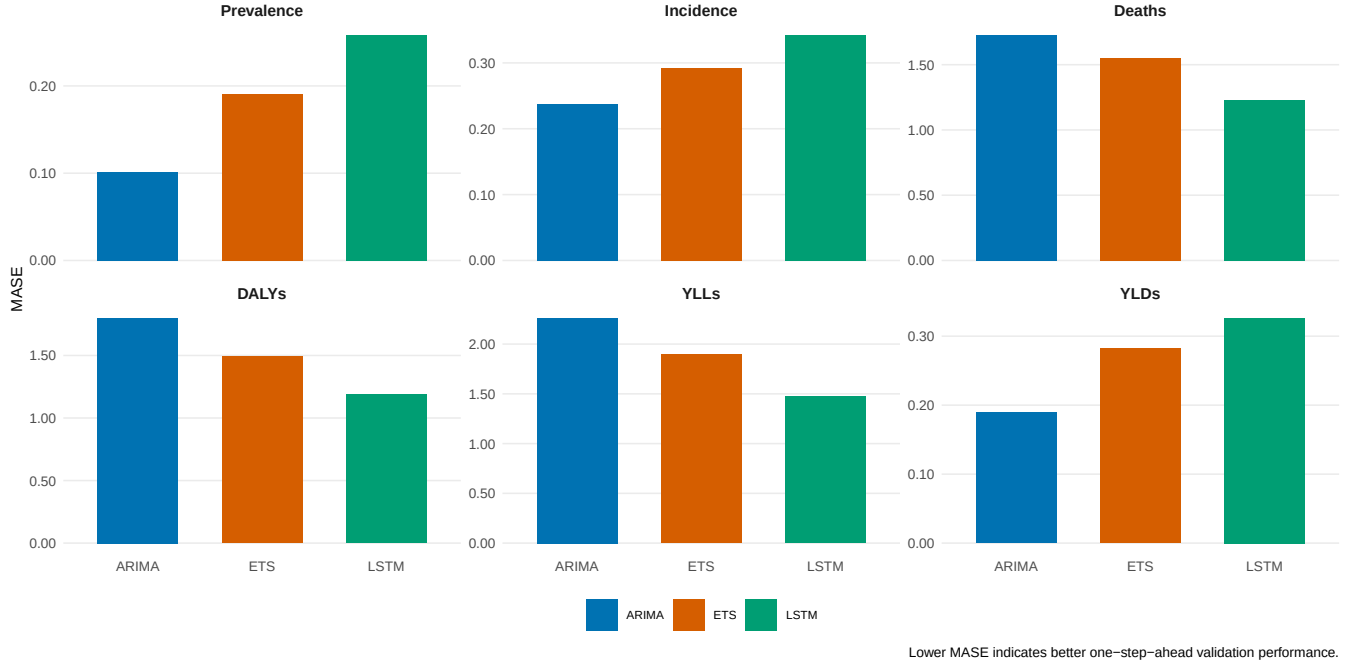


Figure 8: Headline forecast model comparison by MASE.

11 Forecast Validation and 2030 Projections

Table 17: Headline selected-model 2030 forecasts with prediction intervals

Measure	Model	2030 forecast (95% PI)	Interval method
Prevalence	ARIMA	4740.5 (4687.2-4793.8)	ARIMA model prediction interval
Incidence	ARIMA	237.4 (227.3-247.4)	ARIMA model prediction interval
Deaths	LSTM	37.9 (33.4-38.5)	LSTM validation-residual interval
DALYs	LSTM	1296.5 (1199.0-1311.3)	LSTM validation-residual interval
YLLs	LSTM	844.0 (751.2-867.6)	LSTM validation-residual interval
YLDs	ARIMA	437.3 (429.4-445.2)	ARIMA model prediction interval

Notes: This table applies to overall SSA, both sexes, total diabetes. ARIMA and ETS intervals are model prediction intervals. LSTM intervals are empirical validation-residual prediction intervals.

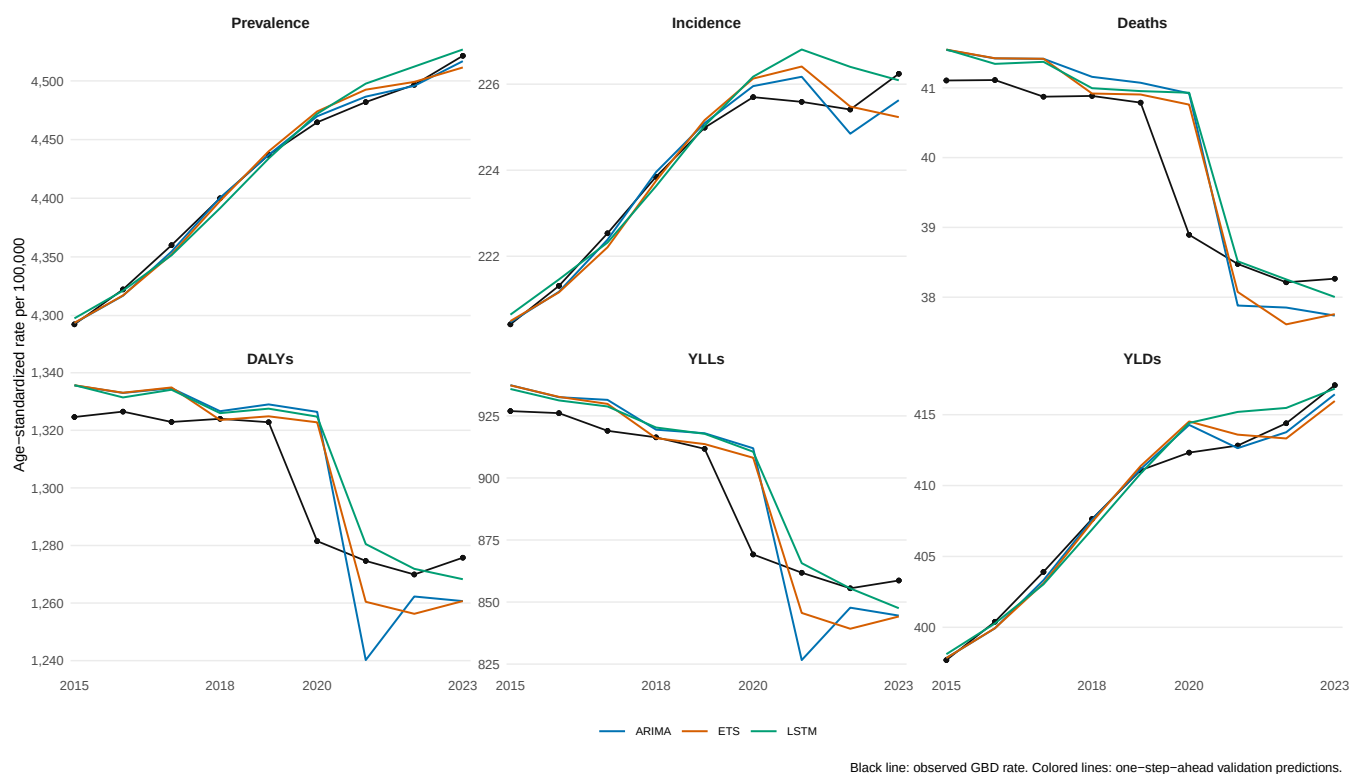


Figure 9: Observed and predicted validation-year rates for headline forecast models.

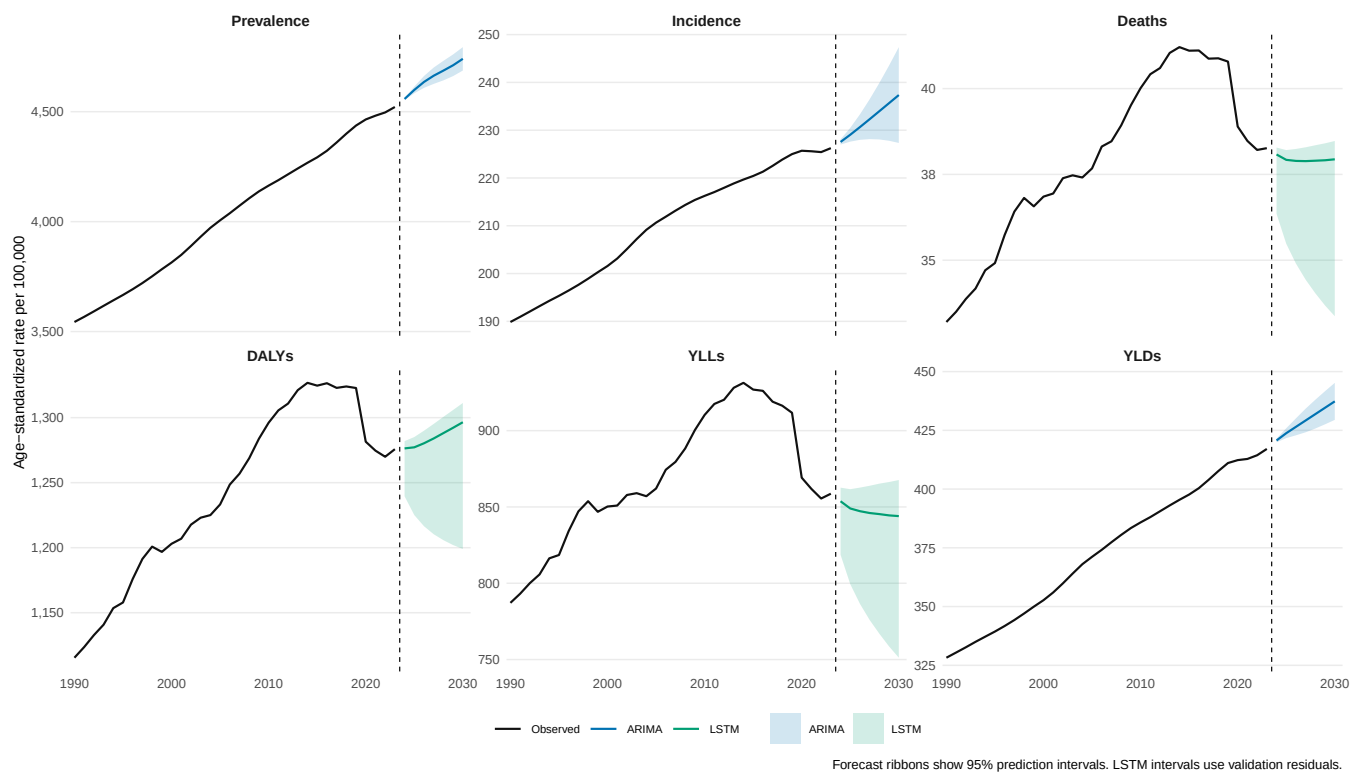


Figure 10: Headline selected-model forecasts through 2030 with prediction intervals.

Table 18: Overall SSA and subregional 2030 prediction intervals

Region	Measure	Model	2030 forecast (95% PI)
SSA	Prevalence	ARIMA	4740.5 (4687.2-4793.8)
SSA	Incidence	ARIMA	237.4 (227.3-247.4)
SSA	Deaths	ETS	36.2 (32.1-40.3)
SSA	DALYs	ETS	1245.3 (1150.6-1340.0)
SSA	YLLs	ETS	814.0 (718.3-909.7)
SSA	YLDs	ARIMA	437.3 (429.4-445.2)
Central SSA	Prevalence	ARIMA	6423.8 (6312.8-6534.9)
Central SSA	Incidence	ARIMA	296.7 (291.1-302.2)
Central SSA	Deaths	ETS	27.1 (22.5-31.7)
Central SSA	DALYs	ARIMA	1315.0 (1232.2-1397.8)
Central SSA	YLLs	ETS	694.5 (596.8-792.2)
Central SSA	YLDs	ARIMA	586.8 (579.9-593.8)
Eastern SSA	Prevalence	ARIMA	3237.2 (3211.3-3263.1)
Eastern SSA	Incidence	ARIMA	177.5 (175.9-179.1)
Eastern SSA	Deaths	ARIMA	37.8 (36.1-39.5)
Eastern SSA	DALYs	ARIMA	1169.1 (1135.2-1203.1)
Eastern SSA	YLLs	ARIMA	797.3 (737.1-857.6)
Eastern SSA	YLDs	ARIMA	316.1 (313.8-318.5)
Southern SSA	Prevalence	ARIMA	6727.5 (6404.1-7050.9)
Southern SSA	Incidence	ARIMA	364.0 (345.5-382.5)
Southern SSA	Deaths	ARIMA	63.7 (45.9-81.5)
Southern SSA	DALYs	ARIMA	2011.9 (1606.6-2417.1)
Southern SSA	YLLs	ARIMA	1385.4 (986.4-1784.4)
Southern SSA	YLDs	ARIMA	622.6 (590.5-654.7)
Western SSA	Prevalence	ARIMA	4905.9 (4843.5-4968.3)
Western SSA	Incidence	ARIMA	233.0 (224.6-241.4)
Western SSA	Deaths	ARIMA	28.4 (25.0-31.9)
Western SSA	DALYs	ETS	1042.8 (972.8-1112.8)
Western SSA	YLLs	ARIMA	618.6 (551.8-685.4)
Western SSA	YLDs	ARIMA	431.0 (425.8-436.1)

Notes: Forecasts are age-standardized rates per 100,000 for both sexes and total diabetes. Subregion forecasts use the full-strata ARIMA/ETS production models. The LSTM-inclusive headline forecast for overall SSA is shown separately above.

The LSTM-inclusive headline forecasts above are the main forecast results for overall SSA, both sexes, total diabetes. Full-strata ARIMA/ETS production outputs are retained as machine-readable supplementary files because they cover every location, sex, measure, and diabetes type. The full production ARIMA and ETS model comparison is saved as `outputs/sub_saharan_africa/full_analysis/tables/forecast_model_accuracy.csv`. The headline ARIMA/ETS/LSTM comparison is saved as `outputs/sub_saharan_africa/full_analysis/tables/headline_forecast_model_comparison.csv`. The headline selected-model forecasts with LSTM prediction intervals are saved as `outputs/sub_saharan_africa/full_analysis/tables/headline_forecast_model_selected.csv`.

12 Age-Specific Count Quality

Table 19: Age-specific count quality summary

Region	Sex	Rows	Measures	Causes	Ages	Years
Central SSA	Both	14484	6	3	25	34
Central SSA	Female	14484	6	3	25	34
Central SSA	Male	14484	6	3	25	34
Eastern SSA	Both	14484	6	3	25	34
Eastern SSA	Female	14484	6	3	25	34
Eastern SSA	Male	14484	6	3	25	34
Southern SSA	Both	14484	6	3	25	34
Southern SSA	Female	14484	6	3	25	34
Southern SSA	Male	14484	6	3	25	34
SSA	Both	14484	6	3	25	34
SSA	Female	14484	6	3	25	34
SSA	Male	14484	6	3	25	34
Western SSA	Both	14484	6	3	25	34
Western SSA	Female	14484	6	3	25	34
Western SSA	Male	14484	6	3	25	34

Notes: Each complete region-sex stratum should contain 34 years, 25 age groups, 6 measures, and 3 diabetes causes.

Age-specific counts, matched GBD age-specific rates, and UN WPP population data are complete. The rate file covers all five regions, three sexes, three causes, six measures, 25 mutually exclusive age groups, and every year from 1990 to 2023.

13 Symmetric Decomposition of Change, 1990-2023

Table 20: Validation of GBD-implied age-sex population denominators

Demographic cells	Cells reconstructed	Maximum relative spread	Median relative spread
12750	12750	1e-07	0

Notes: Population was reconstructed as count/rate x 100,000. Agreement across causes and measures confirms that matched counts and rates use a common denominator.

Table 21: Symmetric decomposition of the change in total diabetes burden, both sexes, 1990-2023

Region	Measure	Observed change	Population size	Age structure	Age-specific rate	Numerical error
SSA	Prevalence	22330618.7	16289788.1	1148522.7	4892307.9	0
SSA	Incidence	1226118.0	931428.7	62347.6	232341.7	0
SSA	Deaths	122389.4	105806.4	4248.7	12334.3	0
SSA	DALYs	5095406.3	4474196.7	172880.7	448329.0	0
SSA	YLLs	3195197.0	3076688.4	82963.7	35544.9	0
SSA	YLDs	1900209.3	1400711.5	86715.1	412782.7	0
Central SSA	Prevalence	3828915.4	2896403.5	174545.9	757966.0	0
Central SSA	Incidence	204168.6	159724.3	8286.9	36157.4	0
Central SSA	Deaths	13326.9	11631.4	636.5	1059.0	0
Central SSA	DALYs	718841.6	615032.4	25954.1	77855.1	0
Central SSA	YLLs	397072.8	370840.1	11528.8	14703.8	0
Central SSA	YLDs	321768.8	244597.0	14019.1	63152.8	0
Eastern SSA	Prevalence	5326467.6	4109043.8	453871.4	763552.4	0
Eastern SSA	Incidence	310532.8	251262.4	24081.4	35189.0	0
Eastern SSA	Deaths	39036.6	38418.8	3799.4	-3181.6	0
Eastern SSA	DALYs	1490764.5	1532969.2	118802.4	-161007.1	0
Eastern SSA	YLLs	1009860.5	1161595.3	78451.1	-230185.9	0
Eastern SSA	YLDs	480904.0	372905.5	38856.0	69142.5	0
Southern SSA	Prevalence	3302885.0	1553384.9	998394.6	751105.5	0
Southern SSA	Incidence	187938.8	91958.3	54500.0	41480.5	0
Southern SSA	Deaths	29204.6	11272.0	8218.8	9713.8	0
Southern SSA	DALYs	983475.9	413157.9	285406.6	284911.4	0
Southern SSA	YLLs	676763.3	272002.2	188485.7	216275.4	0
Southern SSA	YLDs	306712.6	141591.4	96481.2	68640.0	0
Western SSA	Prevalence	9872350.6	7356768.3	-73043.4	2588625.8	0
Western SSA	Incidence	523477.8	401495.5	652.6	121329.7	0
Western SSA	Deaths	40821.3	39736.5	-3075.5	4160.3	0
Western SSA	DALYs	1902324.3	1775429.4	-88879.8	215774.7	0
Western SSA	YLLs	1111500.5	1176222.0	-71078.6	6357.1	0
Western SSA	YLDs	790823.8	600210.9	-18804.7	209417.6	0

Notes: Effects are absolute changes in GBD counts and sum to the reconstructed observed change. Each effect is the average marginal contribution across all six possible factor orderings.

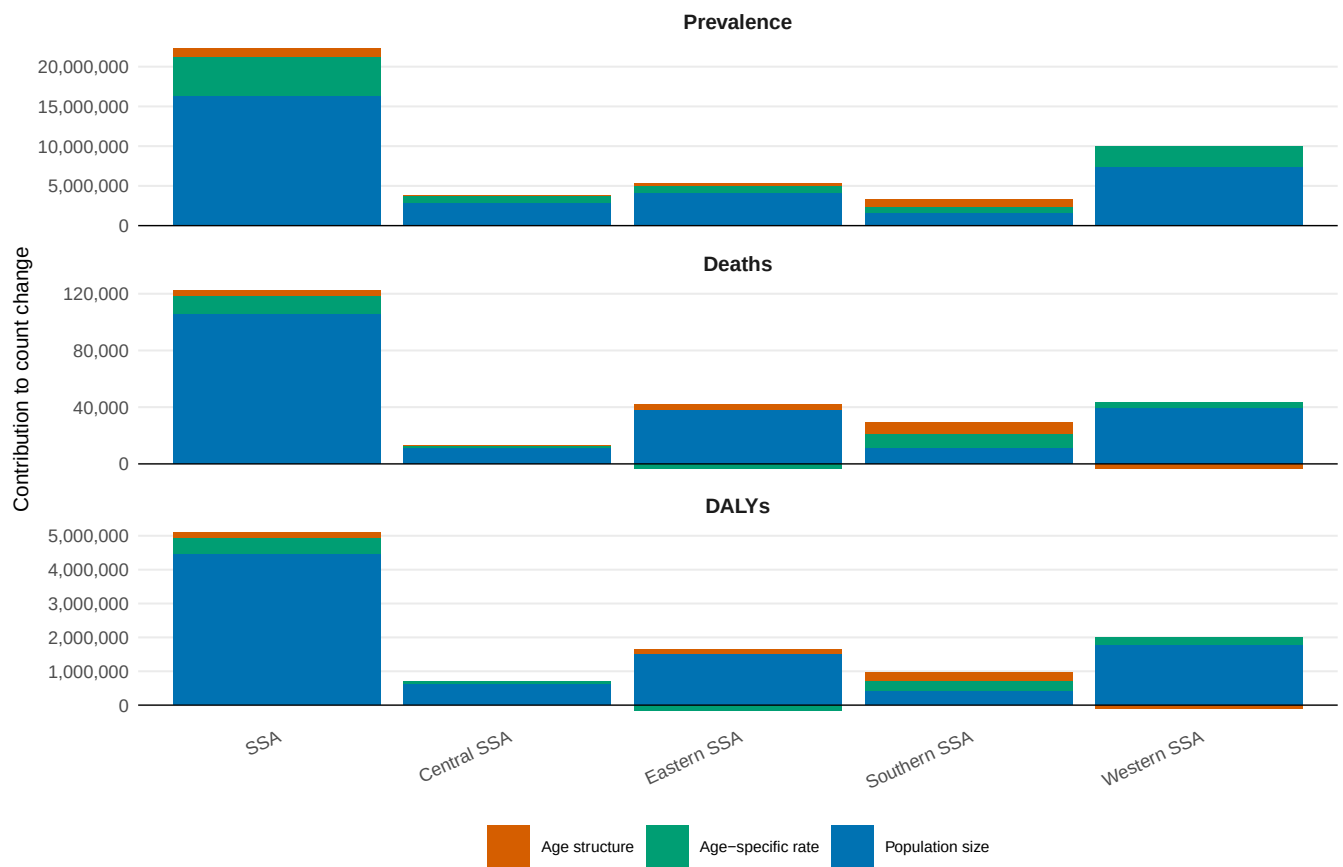


Figure 11: Symmetric decomposition of the change in total diabetes burden, 1990-2023.

14 Age-Specific Patterns and Age-Adjusted Temporal Drift

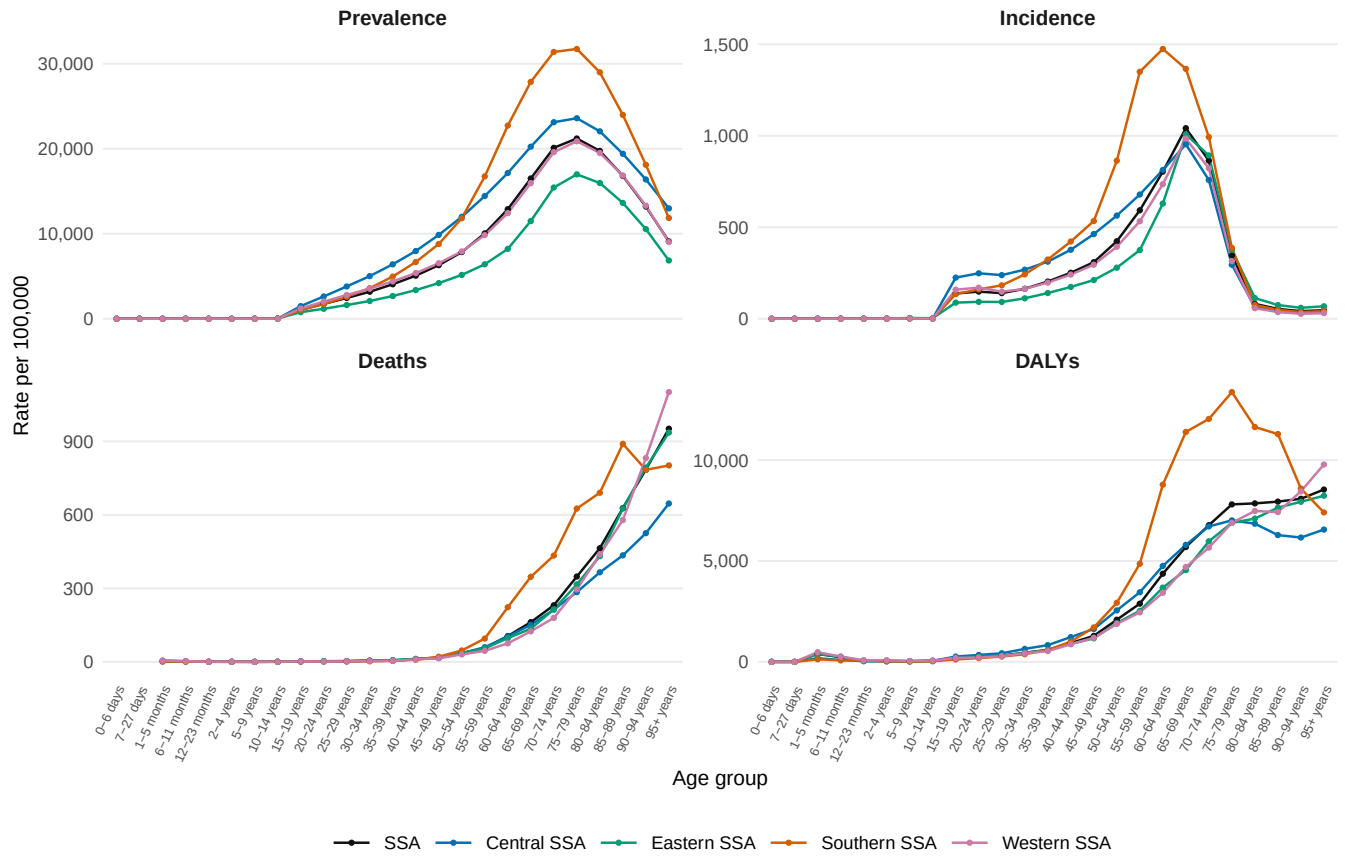


Figure 12: Age-specific total diabetes rates in 2023, both sexes.

Table 22: Age-adjusted annual drift in total diabetes incidence and mortality, 1990-2023

Region	Measure	Annual drift (%)	Lower 95%	Upper 95%	P value
Central SSA	Deaths	0.560	0.516	0.604	0
Central SSA	Incidence	0.612	0.585	0.639	0
Eastern SSA	Deaths	-0.169	-0.217	-0.121	0
Eastern SSA	Incidence	0.355	0.333	0.377	0
Southern SSA	Deaths	1.047	0.959	1.135	0
Southern SSA	Incidence	0.811	0.774	0.849	0
SSA	Deaths	0.340	0.291	0.389	0
SSA	Incidence	0.659	0.634	0.683	0
Western SSA	Deaths	0.404	0.353	0.455	0
Western SSA	Incidence	0.855	0.826	0.884	0

Notes: Quasi-Poisson log-linear models include categorical age effects and GBD-implied population offsets. These estimates quantify age-adjusted temporal drift; they are not presented as a fully identified age-period-cohort decomposition.

15 Population Reconstruction

Table 23: GBD-implied population reconstructed from matched age-specific counts and rates

Region	location_name	Population 1990	Population 2023	Change (%)
Central SSA	Central Sub-Saharan Africa	55208488	146550328	165
Eastern SSA	Eastern Sub-Saharan Africa	190193302	452000600	138
Southern SSA	Southern Sub-Saharan Africa	52748156	88561245	68
SSA	Sub-Saharan Africa	491570793	1206793205	145
Western SSA	Western Sub-Saharan Africa	193420847	519681032	169

Notes: Both-sex population totals are sums of the 25 mutually exclusive age groups.

16 Supplementary Output Catalogue

Table 24: Machine-readable table catalogue

file	directory
age_adjusted_incidence_mortality_drift.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
age_specific_count_quality_summary.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
age_specific_rate_endpoints_1990_2023.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
age_standardized_log_linear_trends.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
age_standardized_rate_endpoints.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
age_standardized_rate_forecasts_2024_2030.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
all_age_count_endpoints.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
diabetes_type_shares.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
forecast_acf_pacf_values.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
forecast_model_accuracy.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
forecast_model_registry.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
forecast_model_selection_summary.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
forecast_stationarity_checks.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
gbd_implied_age_sex_population_1990_2023.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
gbd_implied_population_validation.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
headline_forecast_model_comparison.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
headline_forecast_model_validation_predictions.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
headline_selected_forecast_models.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
headline_selected_model_forecasts_2024_2030.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
input_availability.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
overall_ssa_main_burden_table.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
sex_rate_ratios.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
subregion_2030_prediction_intervals.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
subregion_to_overall_ssa_rate_ratios.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
symmetric_decomposition_1990_2023.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
yll_to_yld_ratio_annual.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,
yll_to_yld_ratio_periods.csv	/Users/christopherdoom/Desktop/Projects_mentor/Diabetes_paperSSA/outputs/sub_saharan_africa,

Notes: CSV tables contain the full machine-readable outputs used to create the compact report tables.

17 Methods Notes

This report uses GBD 2023 point estimates and uncertainty intervals for Sub-Saharan Africa and the four GBD subregions. Age-standardized rates are reported per 100,000 population. Full-period APC estimates come from log-linear models and are descriptive summaries of average annual change. Forecast intervals reflect model uncertainty only; GBD uncertainty in forecasts should be propagated later if draws become available. The symmetric decomposition averages effects across all factor orderings. Age-adjusted drift models use quasi-Poisson regression with categorical age effects and should not be interpreted as a fully identified age-period-cohort decomposition.